

C ERROR ANALYSIS DETAILS

This section provides a detailed quantitative and qualitative breakdown of the error analysis conducted to understand the failure modes of VLMs on Spatial-DISE Bench.

C.1 DEFINITION OF HIGH-LEVEL ERROR

We established a high-level error taxonomy to systematically diagnose failures by deconstructing the model mistakes into three errors:

- **Perceptual Error**, which the model fails to accurately interpret basic visual information, such as the shape, count, or spatial relationship of objects.
- **Comprehension Error**, which the model misinterprets the natural language prompt or the objective of the task, indicating a failure to understand the question.
- **Reasoning Error**, which the model correctly perceives the visual scene and understands the prompt but fails in the logical deduction required to reach the correct answer. This includes errors in mental rotation, folding, or spatial manipulation.

C.2 VLM-AS-JUDGE IN ERROR ANALYSIS

Inspired by Yang et al. (2025), we adopted an automated error analysis pipeline. As shown in Figure 11, we use Doubao-1.6-thinking as a judge, combined with human inspection. Table 15 lists the distribution of the wrong responses sampled in error analysis.

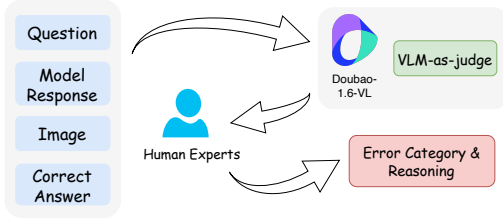


Figure 11: Error Analysis Pipeline

Table 15: Error Distribution by DISE category

DISE Category	Count
Intrinsic-Static (I-S)	34
Intrinsic-Dynamic (I-D)	107
Extrinsic-Static (E-S)	34
Extrinsic-Dynamic (E-D)	25
Total	200

The prompt template used for error analysis:

Listing 6: Prompt Templates used for Error Analysis

```
ERROR_ANALYSIS_PROMPTS = {
    "detailed_analysis": """
Please provide a detailed analysis of the visual-language model's
incorrect answer:

Question Category: {category}
Question: {question}

Options:
{options}

Correct Answer: {correct_answer}
Model's Predicted Answer: {predicted_answer}
Model's Full Response: {model_prediction}

Please analyze in depth from the following perspectives:
1. Error Type Classification:
    - Perception Error: The model failed to correctly identify visual
      elements in the image.
    - Comprehension Error: The model recognized visual elements but
      misunderstood their meaning.
    - Reasoning Error: The model understood the content but made a
      mistake in reasoning.
```

```

2. Specific Cause of the Error
3. Severity Assessment (Low / Medium / High)
4. Possible Directions for Improvement
5. Suggestions to Prevent Similar Errors

Please return the analysis in JSON format:
{
  "Error Type": "Specific type of error",
  "Error Subtype": "More detailed category of the error",
  "Cause of Error": "Detailed explanation of the cause",
  "Severity": "Low/Medium/High",
  "Summary": "Brief summary of the error"
}

"""

    "category_analysis": ""
Please analyze the error patterns of the visual-language model in the
following {category} category questions:

{error_examples}

Analyze from the following perspectives:
1. Most common error types in this category
2. Common features and patterns of errors
3. Category-specific challenges

Please provide a structured response.
"""

    "comparison_analysis": ""
Please compare the error performance of the following models on the same
question:

{model_comparisons}

Analyze:
1. Differences in error types across models
2. Strengths and weaknesses of each model
3. Comparison of error severity

Please provide a detailed comparative analysis.
"""
}

```

Our analysis reveals a clear and consistent pattern: Reasoning Error is the predominant failure category, accounting for an overwhelming 72.5% (145 out of 200) of all analyzed mistakes. Perceptual errors constituted 17.5% of the total, while comprehension errors were the least common at 10%. This distribution strongly suggests that the primary bottleneck for current VLMs is not in visual perception but in complex spatial-logical inference. While this initial classification identifies where the models fail, a more granular analysis is required to understand why they fail.

C.3 A DEEP DIVE INTO REASONING FAILURES

To move from symptom to cause, we performed a deeper analysis of the 145 reasoning errors, re-categorizing them based on the underlying cognitive abilities that are deficient. This approach, inspired by cognitive science, reveals that the models' failures stem from a lack of fundamental cognitive mechanisms for spatial intelligence. We identified three primary root causes.

Failure in Rule Application (44.8%) This was the most critical category of failure. Models demonstrate an ignorance of the fundamental axioms, constraints, and invariances of the geometric world. The errors are not in complex derivations but in the application of basic, non-negotiable

rules. The root cause appears to be a failure to link visual percepts to an abstract library of geometric principles; the models see pixels, not entities governed by rules.

A frequent failure was confusing adjacent and opposite faces in 3D cube problems. For instance, a model might correctly identify the symbols on a cube’s faces but fail to apply the simple rule that adjacent faces cannot be opposite one another.

Failure in Mental Simulation (40.0%) The second most significant failure was the inability to construct a dynamic, operable internal representation to simulate a continuous spatial transformation. Models lack a reliable "spatial working memory" to track an object’s state through a sequence of operations. They cannot robustly answer the question, "what happens next?"

This was most evident in "Fold and Punch" tasks. Models consistently failed to track the number of layers created by folds and, consequently, could not predict the symmetric replication of holes upon unfolding. For example, after simulating a two-fold process (creating four layers), a model might incorrectly predict only two holes in the unfolded paper, demonstrating a breakdown in state tracking.

Failure in Holistic-Local Processing (15.2%) Finally, models exhibited an imbalance in processing visual information, struggling to shift between holistic understanding and local detail analysis. Their attention mechanisms appear unable to dynamically allocate cognitive resources to the most salient features required by the task.

Models were often misled by superficial similarity. In rotation tasks, a model might identify an option as correct simply because it "looks similar" to the target, while ignoring a fatal flaw in the local arrangement of its components, such as an incorrect orientation of a key part.

In summary, the failures of current VLMs in spatial reasoning are systemic and deeply rooted in cognitive deficiencies. They lack an internal "world model" that is constrained by geometric rules, can be manipulated through dynamic simulation in a spatial working memory, and is guided by a flexible attentional mechanism. This points to a clear direction for future research: efforts must transcend simple pattern matching and focus on imbuing models with the foundational capabilities for genuine spatial cognition.

Table 16: Error Analysis Across Different Models

Err. Models	Qwen2.5-VL	GeminiFlash	Doubao-1.5	SpaceThinker
Reasoning Err.	31	31	37	46
Perceptual Err.	12	12	8	3
Comprehension Err.	7	7	5	1